SESSION 21 · FINAL DEVOPS PROJECT

Typeahead, commit to cluster

A search-suggestion app taken through CI/CD, DevSecOps, Kubernetes, Helm, monitoring and GitOps

Vimal Kumar Yadav · 24BCS10273 · October 2026

THE APPLICATION

One app with three jobs

01

Suggest

Ranked prefix suggestions as you type, from an in-memory trie with a Redis cache in front.

02

Manage

A query catalog with create, edit and delete, stored in PostgreSQL with Flyway migrations.

03

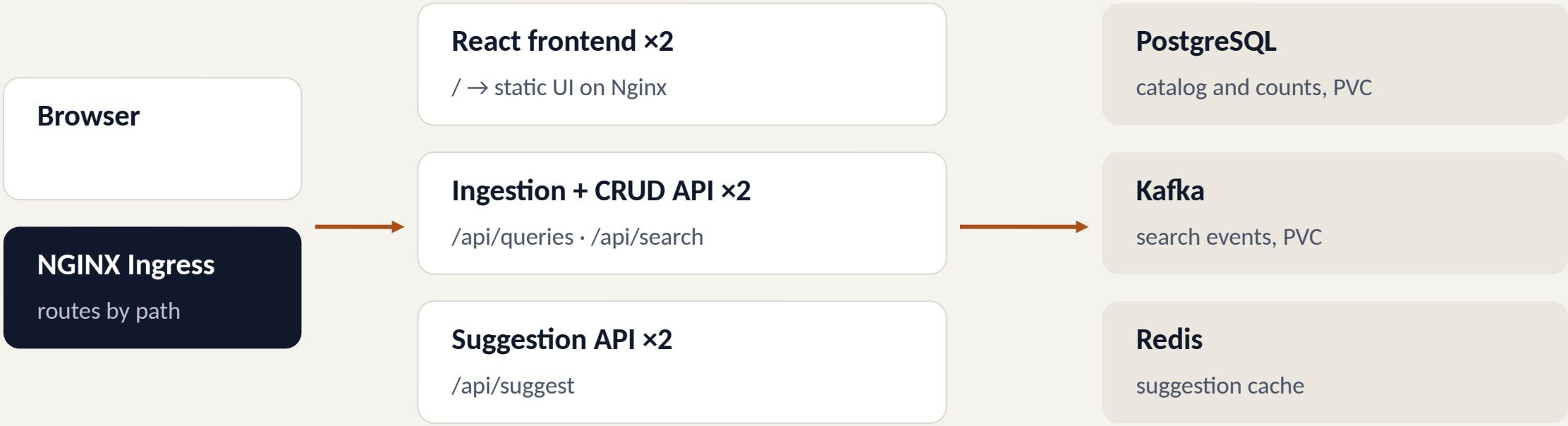
Learn

Each search becomes a Kafka event that raises its count, so suggestions refresh within seconds.

Adapted from my own repository vimalyad/typeahead · Spring Boot, JUnit and Flyway in place of FastAPI, pytest and Alembic (approved)

ARCHITECTURE

Two services, three stores, one Ingress



Images come from GHCR, Argo CD deploys them from Git, and Prometheus scrapes every backend replica for Grafana.

API AND TESTS

Seven routes, fifteen tests

Method	Route	Behaviour
GET	`/api/queries`	List catalog entries
GET	`/api/queries/{id}`	Read one; 404 if absent
POST	`/api/queries`	Create; 201, 409 if duplicate
PUT	`/api/queries/{id}`	Update text and count
DELETE	`/api/queries/{id}`	Delete; 204
POST	`/api/search`	Publish a search event
GET	`/api/suggest?q=`	Ranked suggestions

15

JUnit tests: 12 controller cases, 2 hash-ring tests, 1 concurrency test

Plus a smoke test over real PostgreSQL, Kafka and the trie

CI/CD · GITHUB ACTIONS

Every push runs four gated stages

01

Test

Java tests and the frontend build

02

Scan

Semgrep, dependency gate and secret scan

03

Build

Three non-root images, Trivy gate, push to GHCR by commit SHA

04

Deploy

Helm install on a temporary kind cluster, smoke test, delete

A failed stage stops everything after it, so an image that fails a scan is never published or deployed.

Green release: run 37588149332 · commit 0f8f27c

DEVSECOPS

The security gates blocked two real releases

DEPENDENCY GATE

Critical Spring MVC CVE

No fix existed on Spring 6, so I upgraded to Spring Boot 4.1.1, then patched Tomcat and Jackson.

IMAGE GATE

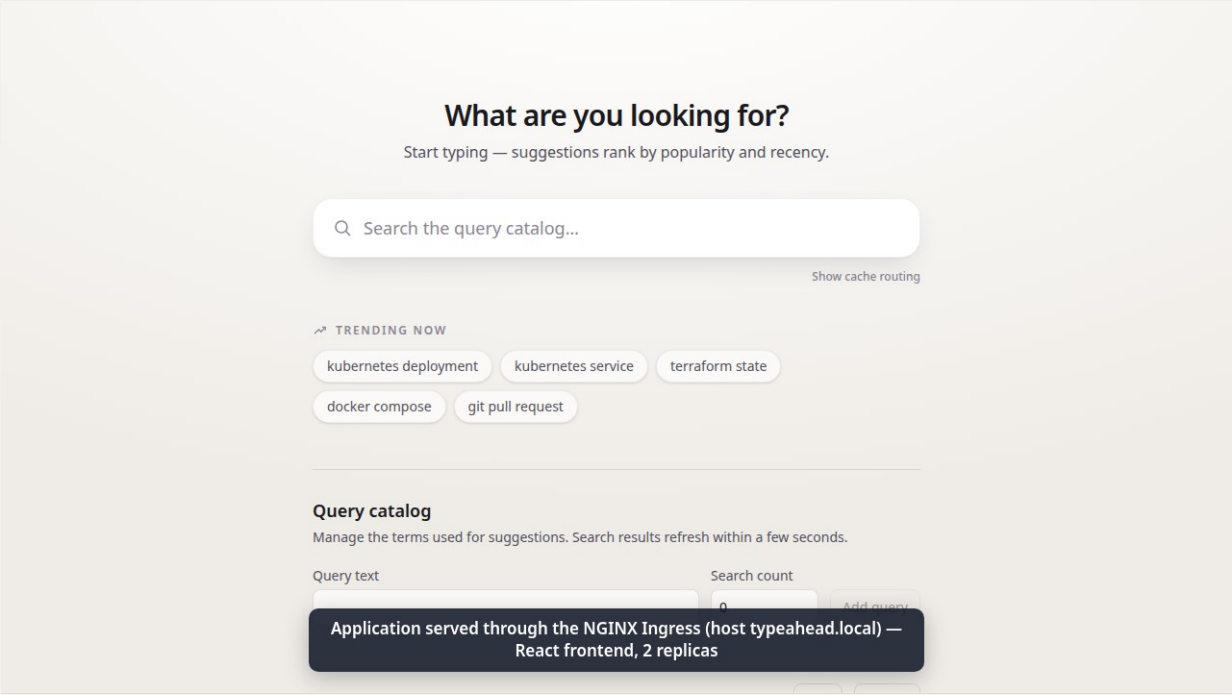
8 HIGH/CRITICAL in the base image

Unfixed Debian libraries, so I switched to the Temurin 21 Alpine JRE and kept the non-root user.

Nothing was suppressed: both fixes passed the same gates.

KUBERNETES + HELM

One Helm chart deploys the whole stack

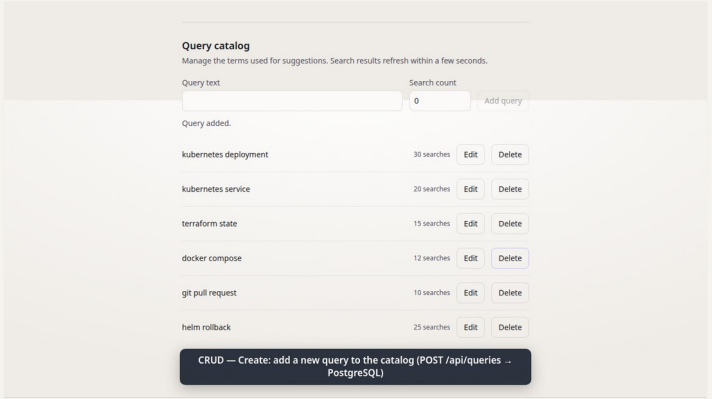


- 2 replicas of each app service
- Startup, readiness and liveness probes
- HPA on CPU, 2 to 3 replicas
- ConfigMap and external Secret
- Ingress for / and /api
- PVCs for PostgreSQL and Kafka
- Non-root, read-only filesystems

The app reached through the NGINX Ingress on local Minikube

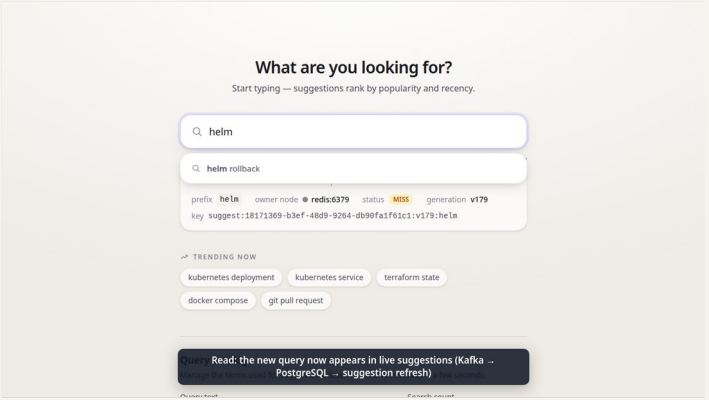
LIVE DEMO

Add a query and it shows up in search



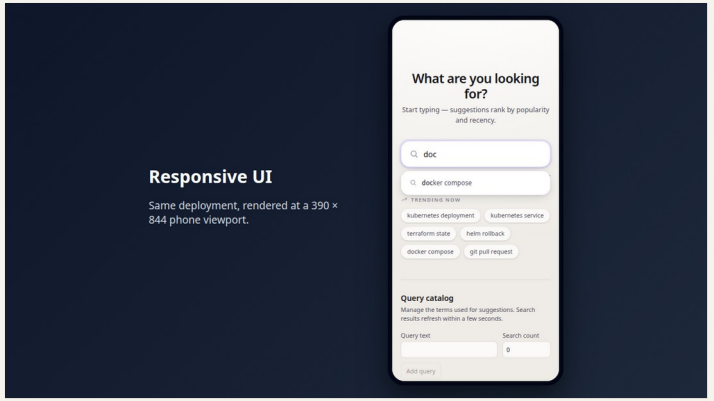
1 · Create

“helm rollback” added through the catalog form, then edited



2 · Suggested

Seconds later it appears as a live suggestion



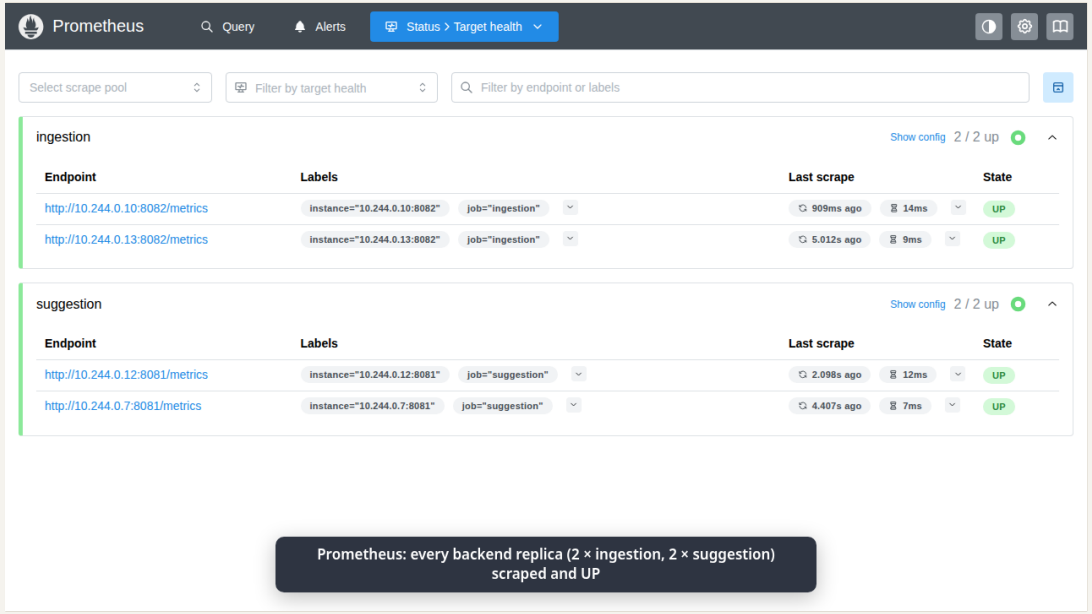
3 · Responsive

Same deployment at a 390 × 844 phone width

Frames from the 90-second browser walkthrough: demo/browser-walkthrough.mp4

MONITORING

Every replica scraped, every request counted



Prometheus: every backend replica (2 × ingestion, 2 × suggestion) scraped and UP



Grafana: live request rate, latency, JVM heap and per-replica scrape health

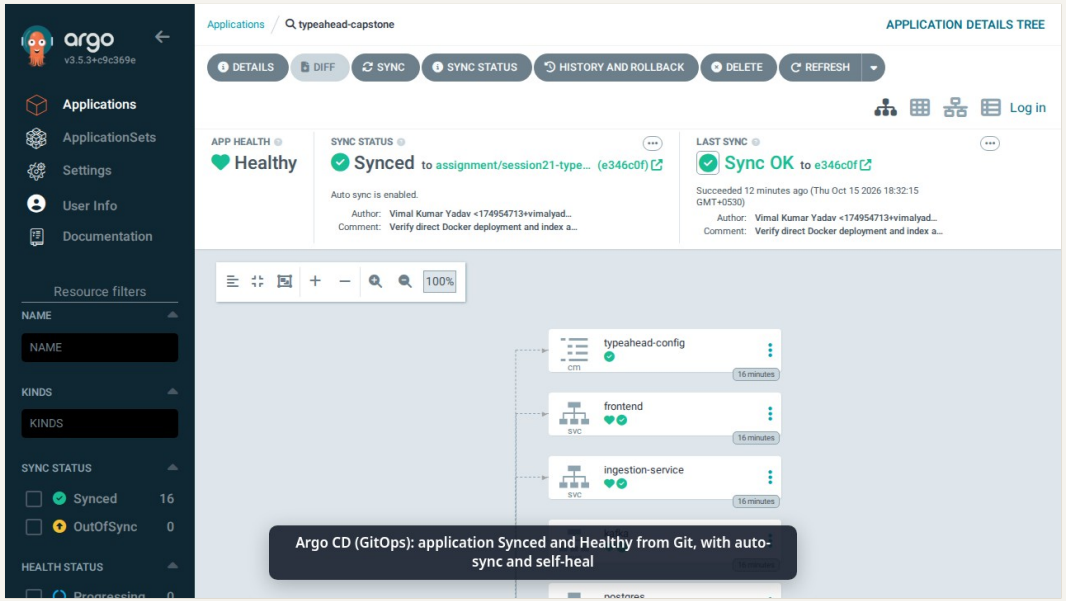
120,415 requests from 16 clients in 80 s with zero failures · the suggestion HPA scaled **2 → 3 → 2**

Left: Prometheus targets, 4 of 4 UP · Right: Grafana dashboard

Git decides what runs, Argo CD enforces it

1. CI publishes images tagged with the commit SHA
2. A promotion commit sets that SHA in gitops/values.yaml
3. Argo CD syncs it automatically, with prune and self-heal
4. The app's /info endpoint reports the same SHA

A hand-edited Service selector was reverted to Git in **2.09 s**



TROUBLESHOOTING CHALLENGE

Four faults, injected and fixed

Fault	What I saw	Root cause and fix	Verified
Kafka DNS missing at startup	No resolvable bootstrap URLs	Worker thread died; added a bounded retry	Recovered with no restart
Wrong Service selector	Ingress 503, no endpoints	Selector matched no pods; restored the label	Same route back to 200
PostgreSQL pod deleted	StatefulSet made a new pod	Checked it mounted the same PVC	Saved row still read 17
Hand edit drifted from Git	Selector changed live	Argo CD self-heal applied Git	Restored in 2.09 s

Each one followed the same loop: identify, check logs and resources, find the root cause, fix, verify.

TERRAFORM · AWS

The cloud design is planned, not applied

What it builds

- VPC with two public subnets
- EKS 1.36 cluster, one managed worker
- Encrypted gp3 storage via Pod Identity
- No NAT gateway or load balancer, to keep cost low

Where it stands

- init and validate pass
- plan: 21 resources to add
- apply and destroy not run: no AWS access
- Real AWS create/destroy done in Sessions 18–19

LESSONS LEARNED

What the project taught me

Readiness can deadlock

Kafka could not resolve its own name until Ready. publishNotReadyAddresses broke the loop.

Background workers must retry

A consumer that gives up on one DNS error silently stops the whole event path.

Scans pay for themselves

Two releases were blocked, and both were fixed at the source instead of ignored.

Tag by SHA, promote by commit

I can always say exactly which build is running and how it got there.

Thank you

Questions?

`github.com/vimalyad/typeahead-devops-capstone`

demo/ holds this deck, its PDF and the 90-second browser walkthrough

README · browser walkthrough video · commit-to-deploy recording

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